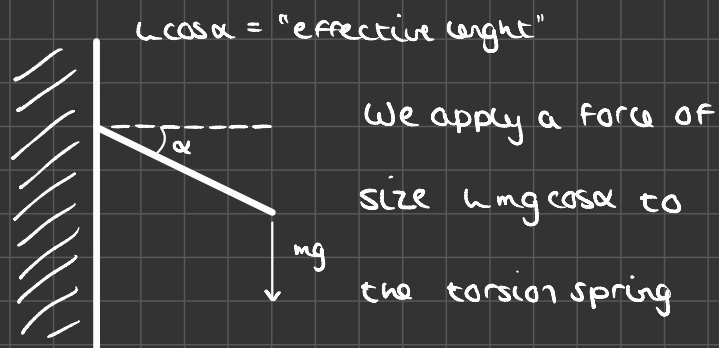
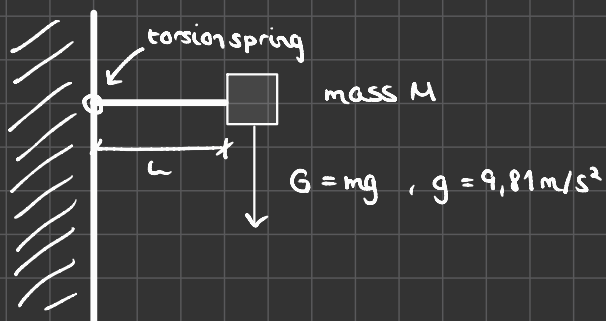


Let's start with an example:



Spring applies ... of size kx , where k is a spring constant

We have an equilibrium of forces when $kx = Lmg \cos \alpha$ or $\alpha = \underbrace{\frac{Lmg}{k}}_{\Theta} \cos \alpha$

We need to solve the equation $\alpha = \Theta \cos \alpha$ with a given $\Theta > 0$

There is no analytical formula to do this, then we need a numerical method

General situation:

We have a function $f: \mathbb{R} \rightarrow \mathbb{R}$ and we want to solve $f(x) = 0$

[e.g.: $f(x) = x - \alpha \cos x$]

Fixed point iterations:

starting point: we write the eq. in fixed point form $g(x) = x$

E.g.:

$$g(x) = x + f(x)$$

$$g(x) = x - f(x)$$

[e.g.: $f(x) = x - \alpha \cos x$]

Fixed point iterations:

We apply g repeatedly until "nothing happens"

For $k = 0, 1, 2, \dots$

$$x_{k+1} = g(x_k)$$

Ex. $\alpha = \frac{1}{2}$

$$g(x) = \frac{1}{2} \cos(x)$$

$$x_0 = 0$$

$$x_1 = \frac{1}{2} \cos(x_0) = \frac{1}{2}$$

$$x_2 = \frac{1}{2} \cos(1/2)$$

Theorem:

Assume that $g: \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable and let $[a, b] \in \mathbb{R}$ be some interval. Assume that $g(x) \in [a, b]$ if $x \in [a, b]$. Then there exists $0 < L < 1$ such that $|g'(x)| \leq L$ for all $x \in [a, b]$. Then the eq. $g(x) = x$ has a unique solution in $[a, b]$. Then for all $x \in [a, b]$ the ite $x_{k+1} = g(x_k)$ converges to the solution r of the eq. $g(x) = x$. We have the following estimates

$$|x_{k+1} - r| \leq L |x_k - r| \leq \frac{L}{1-L} |x_{k+1} - x_k|$$

Proof ideas:

We have that $g(x) \in [a, b] \quad \forall x \in [a, b]$

$$\begin{aligned} g(a) &\geq a & \Rightarrow & g(a) - a \geq 0 \\ g(b) &\leq b & \Rightarrow & g(b) - b \leq 0 \end{aligned}$$

Then there exist $x \in [a, b]$ s.t. $g(x) - x = 0 \Rightarrow g(x) = x$

Now let $x, y \in [a, b]$

Then

$$|g(y) - g(x)| = \left| \int_x^y g'(s) ds \right| \leq \int_x^y \underbrace{|g'(s)|}_{\leq L} ds \leq \int_x^y L ds = L|y-x|$$

So

$$|g(y) - g(x)| \leq L|y-x|$$

with $y = x_k$ and $x = r$ we get

$$|x_{k+1} - r| = |g(x_k) - g(r)| \leq L|x_k - r| \leq L^2|x_{k-1} - r| \leq L^3|x_{k-2} - r| \dots$$

$\nearrow L \rightarrow 0$ as $k \rightarrow \infty$ since $L < 1$

Ex 1:

$$g(x) = \frac{1}{2} \cos(x) \quad 0 \leq x \leq \frac{\pi}{2}$$

We know that $g(x)$ is increasing throughout the interval $g(0) \leq \frac{\pi}{2}$, $g(\frac{\pi}{2}) \leq 0$

$g(a) \in [a, b]$, $g(b) \in [a, b]$ g is decreasing $\Rightarrow g(x) \in [a, b]$ when $x \in [a, b]$

We have $g'(x) = -\frac{1}{2} \sin(x) \Rightarrow |g'(x)| \leq \frac{1}{2} = L < 1$

Fixed point thm is applicable

We have the estimate

$$|x_k - r| \leq \frac{L^k}{1-L} |x_1 - x_0|$$

there $x_0 = 0$, $x_1 = \frac{1}{2}$, $L = \frac{1}{2}$

$$|x_k - r| \leq \frac{(\frac{1}{2})^k}{1 - \frac{1}{2}} \cdot \left| \frac{1}{2} - 0 \right| = \frac{1}{2^k}$$

then we know that the error of the 10th step is at most

$$|x_{10} - r| \leq \frac{1}{2^{10}}$$

How many steps do we need until

$$|x_k - r| \leq 10^{-6}$$

We know that the error is

$$|x_k - r| \leq \frac{1}{2^k}$$

$$k \geq \log_{10}(10^6)$$

$$\frac{1}{2^k} \leq 10^{-6} \Rightarrow 2^k \geq 10^6 \Rightarrow k \geq \frac{6 \log 10}{\log 2}$$

Remark:

One can show if r is a solution of $g(x)=x$ and $|g'(r)| < 1$, then the FPI converges if we start close enough to r .

Reason: if x_0 is close to r , then $\underbrace{|g(x_0) - g(r)|}_{|x_1 - r|} \approx \underbrace{|g'(r)|}_{< 1} |x_0 - r|$

But if $|g'(r)| > 1$ then $|x_1 - r| = |g(x_0) - g(r)| \approx |g'(r)| |x_0 - r| > |x_0 - r|$

$$g(x) = 20 \cos(x)$$

$$\leadsto r \text{ of } g(x) = x \sim \pi/2 \leadsto g'(r) = -20 \sin r \approx 20$$